3.2 Conditional Probabilities

Suppose that we toss 2 dice, and suppose that each of the 36 possible outcomes is equally likely to occur and hence has probability 136. Suppose further that we observe that the first die is a 3. Then, given this information, what is the probability that the sum of the 2 dice equals 8? To calculate this probability, we reason as follows: Given that the initial die is a 3, there can be at most 6 possible outcomes of our experiment, namely, (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6). Since each of these outcomes originally had the same probability of occurring, the outcomes should still have equal probabilities. That is, given that the first die is a 3, the (conditional) probability of each of the outcomes (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6) is 16, whereas the (conditional) probability of the other 30 points in the sample space is 0. Hence, the desired probability will be 16.

If we let E and F denote, respectively, the event that the sum of the dice is 8 and the event that the first die is a 3, then the probability just obtained is called the *conditional probability that E occurs given that F has occurred* and is denoted by

$P(E|F)$

A general formula for $P(E|F)$ that is valid for all events E and F is derived in the same manner: If the event F occurs, then, in order for E to occur, it is necessary that the actual occurrence be a point both in E and in F; that is, it must be in EF. Now, since we know that F has occurred, it follows that F becomes our new, or reduced, sample space; hence, the probability that the event EF occurs will equal the probability of *EF* relative to the probability of F. That is, we have the following definition.

> Definition
>
> If $P(F)>0$, then
>
> $$P(E|F)=P(EF)P(F) \tag{2.1}$$

Example 2a

Joe is 80 percent certain that his missing key is in one of the two pockets of his hanging jacket, being 40 percent certain it is in the left-hand pocket and 40 percent certain it is in the right-hand pocket. If a search of the left-hand pocket does not find the key, what is the conditional probability that it is in the other pocket?

Solution

If we let L be the event that the key is in the left-hand pocket of the jacket, and R be the event that it is in the right-hand pocket, then the desired probability $P(R|Lc)$ can be obtained as follows:

$$P(R|Lc)=P(RLc)P(Lc)=P(R)1−P(L)=2/3$$

If each outcome of a finite sample space S is equally likely, then, conditional on the event that the outcome lies in a subset $F⊂S$, all outcomes in F become equally likely. In such cases, it is often convenient to compute conditional probabilities of the form $P(E|F)$ by using F as the sample space. Indeed, working with this reduced sample space often results in an easier and better understood solution. Our next two examples illustrate this point.

Example 2b

A coin is flipped twice. Assuming that all four points in the sample space $S={(h,h),(h,t),(t,h),(t,t)}$ are equally likely, what is the conditional probability that both flips land on heads, given that (a) the first flip lands on heads? (b) at least one flip lands on heads?

Solution

Let $B={(h,h)}$ be the event that both flips land on heads; let $F={(h,h),(h,t)}$ be the event that the first flip lands on heads; and let $A={(h,h),(h,t),(t,h)}$ be the event that at least one flip lands on heads. The probability for (a) can be obtained from

$$P(B|F)=P(BF)P(F)=P({(h,h)})P({(h,h),(h,t)})=1/42/4=1/2$$

For (b), we have

$$P(B|A)=P(BA)P(A)=P({(h,h)})P({(h,h),(h,t),(t,h)})=1/43/4=1/3$$

Thus, the conditional probabilityhat both flips land on heads given that the first one does is 1/2, whereas the conditional probability that both flips land on heads given that at least one does is only 1/3. Many students initially find this latter result surprising. They reason that given that at least one flip lands on heads, there are two possible results: Either they both land on heads or only one does. Their mistake, however, is in assuming that these two possibilities are equally likely. Initially there are 4 equally likely outcomes. Because the information that at least one flip lands on heads is equivalent to the information that the outcome is not (t,t), we are left with the 3 equally likely outcomes (h,h),(h,t),(t,h), only one of which results in both flips landing on heads.

Example 2c

In the card game bridge, the 52 cards are dealt out equally to 4 players–called East, West, North, and South. If North and South have a total of 8 spades among them, what is the probability that East has 3 of the remaining 5 spades?

Solution

Probably the easiest way to compute the desired probability is to work with the reduced sample space. That is, given that North–South have a total of 8 spades among their 26 cards, there remains a total of 26 cards, exactly 5 of them being spades, to be distributed among the East–West hands. Since each distribution is equally likely, it follows that the conditional probability that East will have exactly 3 spades among his or her 13 cards is

$$(53)(2110)(2613)≈.339$$

Multiplying both sides of **Equation (2.1)** by P(F), we obtain

$$P(EF)=P(F)P(E|F) \tag{2.2}$$

In words, **Equation (2.2)** states that the probability that both E and F occur is equal to the probability that F occurs multiplied by the conditional probability of E given that F occurred. **Equation (2.2)** is often quite useful in computing the probability of the intersection of events.

Example 2d

Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be 12 in a French course and 23 in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

Solution

Let *C* be the event that Celine takes chemistry and A denote the event that she receives an A in whatever course she takes, then the desired probability is $P(CA)$, which is calculated by using **Equation (2.2)** as follows:

$$P(CA)=P(C)P(A|C)=(12)(23)=13$$

Example 2e

Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement. (a) If we assume that at each draw, each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red? (b) Now suppose that the balls have different weights, with each red ball having weight r and each white ball having weight w. Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?

Solution

Let R1 and R2 denote, respectively, the events that the first and second balls drawn are red. Now, given that the first ball selected is red, there are 7 remaining red balls and 4 white balls, so $P(R2|R1)=711$. As $P(R1)$ is clearly 812, the desired probability is

P(R1R2)=P(R1)P(R2|R1)=(23)(711)=1433

Of course, this probability could have been computed by P(R1R2)=(82)/(122).

For part (b), we again let Ri be the event that the ith ball chosen is red and use

P(R1R2)=P(R1)P(R2|R1)

Now, number the red balls, and let Bi,i=1,…,8 be the event that the first ball drawn is red ball number i. Then

P(R1)=P(∪i=18Bi)=∑i=18P(Bi)=8 r8r+4w

Moreover, given that the first ball is red, the urn then contains 7 red and 4 white balls. Thus, by an argument similar to the preceding one,

P(R2|R1)=7r7r+4w

Hence, the probability that both balls are red is

P(R1R2)=8r8r+4w 7r7r+4w

A generalization of **Equation (2.2)**, which provides an expression for the probability of the intersection of an arbitrary number of events, is sometimes referred to as the *multiplication rule*.

> The multiplication rule
>
> P(E1E2E3⋯En)=P(E1)P(E2|E1)P(E3|E1E2)⋯P(En|E1⋯En−1)

In words, the multiplication rule states that P(E1E2⋯En), the probability that all of the events E1,E2,…,En occur, is equal to P(E1), the probability that E1 occurs, multiplied by P(E2|E1), the conditional probability that E2 occurs given that E1 has occurred, multiplied by P(E3|E1E2), the conditional probability that E3 occurs given that both E1 and E2 have occurred, and so on.

To prove the multiplication rule, just apply the definition of conditional probability to its right-hand side, giving

P(E1)P(E1E2)P(E1)P(E1E2E3)P(E1E2)⋯P(E1E2⋯En)P(E1E2⋯En−1)=P(E1E2⋯En)

Example 2f

In the match problem stated in **Example 5m** of **Chapter 2**, it was shown that PN, the probability that there are no matches when N people randomly select from among their own N hats, is given by

PN=∑i=0N(−1)i/i!

What is the probability that exactly k of the N people have matches?

Solution

Let us fix our attention on a particular set of k people and determine the probability that these k individuals have matches and no one else does. Letting E denote the event that everyone in this set has a match, and letting G be the event that none of the other N−k people have a match, we have

P(EG)=P(E)P(G|E)

Now, let Fi,i=1,…,k, be the event that the ith member of the set has a match. Then

P(E)=P(F1F2⋯Fk)=P(F1)P(F2|F1)P(F3|F1F2)⋯P(Fk|F1⋯Fk−1)=1N 1N−1 1N−2⋯1N−k+1=(N−k)!N!

Given that everyone in the set of k has a match, the other N−k people will be randomly choosing among their own N−k hats, so the probability that none of them has a match is equal to the probability of no matches in a problem having N−k people choosing among their own N−k hats. Therefore,

P(G|E)=PN−k=∑i=0N−k(−1)i/i!

showing that the probability that a specified set of k people have matches and no one else does is

P(EG)=(N−k)!N! PN−k

Because there will be exactly k matches if the preceding is true for any of the (Nk) sets of k individuals, the desired probability is

P(exactly k matches)=(Nk)P(EG)=PN−K/K!≈e−1/k!when N is large

Example 2g

An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

Solution

Define events Ei,i=1,2,3,4, as follows:

E1={the ace of spades is in any one of the piles}E2={the ace of spades and the ace of hearts are in different piles}E3={the aces of spades, hearts, and diamonds are all in different piles}E4={all 4 aces are in different piles}

The desired probability is P(E1E2E3E4), and by the multiplication rule,

P(E1E2E3E4)=P(E1)P(E2|E1)P(E3|E1E2)P(E4|E1E2E3)

Now,

P(E1)=1

since E1 is the sample space S. To determine P(E2|E1), consider the pile that contains the ace of spades. Because its remaining 12 cards are equally likely to be any 12 of the remaining 51 cards, the probability that the ace of hearts is among them is 12/51, giving that

P(E2|E1)=1−1251=3951

Also, given that the ace of spades and ace of hearts are in different piles, it follows that the set of the remaining 24 cards of these two piles is equally likely to be any set of 24 of the remaining 50 cards. As the probability that the ace of diamonds is one of these 24 is 24/50, we see that

P(E3|E1E2)=1−2450=2650

Because the same logic as used in the preceding yields that

P(E4|E1E2E3)=1−3649=1349

the probability that each pile has exactly 1 ace is

P(E1E2E3E4)=39·26·1351·50·49≈.105

That is, there is approximately a 10.5 percent chance that each pile will contain an ace. (**Problem 13** gives another way of using the multiplication rule to solve this problem.)

Example 2h

Four of the eight teams in the quarterfinal round of the 2016 European Champions League Football (soccer) tournament were the acknowledged strong teams Barcelona, Bayern Munich, Real Madrid, and Paris St-Germain. The pairings in this round are supposed to be totally random, in the sense that all possible pairings are equally likely. Assuming this is so, find the probability that none of the strong teams play each other in this round. (Surprisingly, it seems to be a common occurrence in this tournament that, even though the pairings are supposedly random, the very strong teams are rarely matched against each other in this round.)

Solution

If we number the four strong teams 1 through 4, and then let Wi,i=1,2,3,4, be the event that team i plays one of the four weak teams, then the desired probability is P(W1W2W3W4). By the multiplication rule

P(W1W2W3W4)=P(W1)P(W2|W1)P(W3|W1W2)P(W4|W1W2W3)=(4/7)(3/5)(2/3)(1)=8/35

The preceding follows by first noting that because team 1 is equally likely to be matched with any of the other 7 teams, we have that P(W1)=4/7. Now, given that W1 occurs, team 2 is equally likely to be matched with any of five teams: namely, teams 3, 4, or any of the three weak teams not matched with team 1. As three of these five teams are weak, we see that P(W2|W1)=3/5. Similarly, given that events W1 and W2 have occurred, team 3 is equally likely to be matched with any from a set of three teams, consisting of team 4 and the remaining two weaker teams not matched with 1 or 2. Hence, P(W3|W1W2)=2/3. Finally, given that W1, W2, and W3 all occur, team 4 will be matched with the remaining weak team not matched with any of 1,2,3, giving that P(W4|W1W2W3)=1.

Remarks Our definition of P(E|F) is consistent with the interpretation of probability as being a long-run relative frequency. To see this, suppose that n repetitions of the experiment are to be performed, where n is large. We claim that if we consider only those experiments in which F occurs, then P(E|F) will equal the long-run proportion of them in which E also occurs. To verify this statement, note that since P(F) is the long-run proportion of experiments in which F occurs, it follows that in the n repetitions of the experiment, F will occur approximately nP(F) times. Similarly, in approximately nP(EF) of these experiments, both E and F will occur. Hence, out of the approximately nP(F) experiments in which F occurs, the proportion of them in which E also occurs is approximately equal to

nP(EF)nP(F)=P(EF)P(F)

Because this approximation becomes exact as n becomes larger and larger, we have the appropriate definition of P(E|F).

3.3 Bayes's Formula

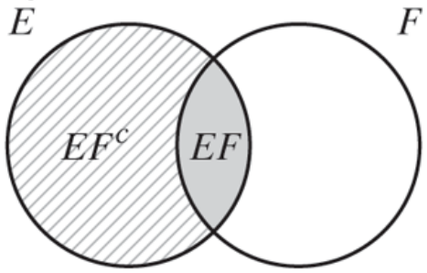
Let E and F be events. We may express E as

$$E=EF\cup EFc$$

for, in order for an outcome to be in E, it must either be in both E and F or be in E but not in F. (See **Figure 3.1**.) As *EF* and EFc are clearly mutually exclusive, we have, by Axiom 3,

$$P(E)=P(EF)+P(EFc)=P(E|F)P(F)+P(E|Fc)P(Fc)=P(E|F)P(F)+P(E|Fc)[1-P(F)] \quad (3.1)$$

Figure 3.1 E=EF∪EFc. EF=Shaded Area; EFc=Striped Area.



Equation (3.1) states that the probability of the event E is a weighted average of the conditional probability of E given that F has occurred and the conditional probability of E given that F has not occurred—each conditional probability being given as much weight as the event on which it is conditioned has of occurring. This is an extremely useful formula, because its use often enables us to determine the probability of an event by first "conditioning" upon whether or not some second event has occurred. That is, there are many instances in which it is difficult to compute the probability of an event directly, but it is straightforward to compute it once we know whether or not some second event has occurred. We illustrate this idea with some examples.

Example 3a

(Part 1)

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a person who is not accident prone. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

Solution

We shall obtain the desired probability by first conditioning upon whether or not the policyholder is accident prone. Let A1 denote the event that the policyholder will have an accident within a year of purchasing the policy, and let A denote the event that the policyholder is accident prone. Hence, the desired probability is given by

$$P(A1)=P(A1|A)P(A)+P(A1|Ac)P(Ac)=(.4)(.3)+(.2)(.7)=.26$$

Example 3a

(Part 2)

Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

Solution

The desired probability is

$$P(A|A1)=P(AA1)P(A1)=P(A)P(A1|A)P(A1)=(.3)(.4).26=613$$

Example 3b

Consider the following game played with an ordinary deck of 52 playing cards: The cards are shuffled and then turned over one at a time. At any time, the player can guess that the next card to be turned over will be the ace of spades; if it is, then the player wins. In addition, the player is said to win if the ace of spades has not yet appeared when only one card remains and no guess has yet been made. What is a good strategy? What is a bad strategy?

Solution

Every strategy has probability 1/52 of winning! To show this, we will use induction to prove the stronger result that for an n card deck, one of whose cards is the ace of spades, the probability of winning is 1/n, no matter what strategy is employed. Since this is clearly true for n=1, assume it to be true for an n−1 card deck, and now consider an n card deck. Fix any strategy, and let p denote the probability that the strategy guesses that the first card is the ace of spades. Given that it does, the player's probability of winning is 1/n. If, however, the strategy does not guess that the first card is the ace of spades, then the probability that the player wins is the probability that the first card is not the ace of spades, namely, (n−1)/n, multiplied by the conditional probability of winning given that the first card is not the ace of spades. But this latter conditional probability is equal to the probability of winning when using an n−1 card deck containing a single ace of spades; it is thus, by the induction hypothesis, 1/(n−1). Hence, given that the strategy does not guess the first card, the probability of winning is

$$\frac{n-1}{n}\frac{1}{n-1}=\frac{1}{n}$$

Thus, letting G be the event that the first card is guessed, we obtain

$$P\{\text{win}\}=P\{\text{win}|G\}P(G)+P\{\text{win}|G^c\}(1-P(G))=\frac{1}{n}p+\frac{1}{n}(1-p)=\frac{1}{n}$$

Example 3c

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and 1−p be the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability 1/m, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

Solution

Let C and K denote, respectively, the events that the student answers the question correctly and the event that he or she actually knows the answer. Now,

$$P(K|C)=\frac{P(KC)}{P(C)}=\frac{P(C|K)P(K)}{P(C|K)P(K)+P(C|K^c)P(K^c)}=\frac{p}{p+(1/m)(1-p)}=\frac{mp}{1+(m-1)p}$$

For example, if m=5,p=12, then the probability that the student knew the answer to a question he or she answered correctly is 56.

Example 3d

A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Solution

Let D be the event that the person tested has the disease and E the event that the test result is positive. Then the desired probability is

$$P(D|E)=\frac{P(DE)}{P(E)}=\frac{P(E|D)P(D)}{P(E|D)P(D)+P(E|D^c)P(D^c)}=\frac{(.95)(.005)}{(.95)(.005)+(.01)(.995)}=\frac{95}{294}\approx.323$$

Thus, only 32 percent of those persons whose test results are positive actually have the disease. Many students are often surprised at this result (they expect the percentage to be much higher, since the blood test seems to be a good one), so it is probably worthwhile to present a second argument that, although less rigorous than the preceding one, is probably more revealing. We now do so.

Since .5 percent of the population actually has the disease, it follows that, on the average, 1 person out of every 200 tested will have it. The test will correctly confirm that this person has the disease with probability .95. Thus, on the average, out of every 200 persons tested, the test will correctly confirm that .95 person has the disease. On the other hand, out of the (on the average) 199 healthy people, the test will incorrectly state that (199)(.01) of these people have the disease. Hence, for every .95 diseased persons that the test correctly states is ill, there are (on the average) (199)(.01) healthy persons who the test incorrectly states are ill. Thus, the proportion of time that the test result is correct when it states that a person is ill is

$$\frac{.95}{.95+(199)(.01)}=\frac{95}{294}\approx.323$$

Equation (3.1) is also useful when one has to reassess one's personal probabilities in the light of additional information. For instance, consider the examples that follow.

Example 3e

Consider a medical practitioner pondering the following dilemma: "If I'm at least 80 percent certain that my patient has this disease, then I always recommend surgery, whereas if I'm not quite as certain, then I recommend additional tests that are expensive and sometimes painful. Now, initially I was only 60 percent certain that Jones had the disease, so I ordered the series A test, which always gives a positive result when the patient has the disease and almost never does when he is healthy. The test result was positive, and I was all set to

recommend surgery when Jones informed me, for the first time, that he was diabetic. This information complicates matters because, although it doesn't change my original 60 percent estimate of his chances of having the disease in question, it does affect the interpretation of the results of the A test. This is so because the A test, while never yielding a positive result when the patient is healthy, does unfortunately yield a positive result 30 percent of the time in the case of *diabetic* patients who are not suffering from the disease. Now what do I do? More tests or immediate surgery?"

Solution

In order to decide whether or not to recommend surgery, the doctor should first compute her updated probability that Jones has the disease given that the A test result was positive. Let D denote the event that Jones has the disease and E the event that the A test result is positive. The desired conditional probability is then

$$P(D|E)=P(DE)P(E)=P(D)P(E|D)P(E|D)P(D)+P(E|Dc)P(Dc)=(.6)11(.6)+(.3)(.4)=.833$$

Note that we have computed the probability of a positive test result by conditioning on whether or not Jones has the disease and then using the fact that because Jones is a diabetic, his conditional probability of a positive result given that he does not have the disease, P(E|Dc), equals .3. Hence, as the doctor should now be more than 80 percent certain that Jones has the disease, she should recommend surgery.

Example 3f

At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect. Suppose, however, that a *new* piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered. If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?

Solution

Letting G denote the event that the suspect is guilty and C the event that he possesses the characteristic of the criminal, we have

$$P(G|C)=P(GC)P(C)=P(C|G)P(G)P(C|G)P(G)+P(C|Gc)P(Gc)=1(.6)1(.6)+(.2)(.4)\approx.882$$

where we have supposed that the probability of the suspect having the characteristic if he is, in fact, innocent is equal to .2, the proportion of the population possessing the characteristic.

Example 3g

In the world bridge championships held in Buenos Aires in May 1965, the famous British bridge partnership of Terrence Reese and Boris Schapiro was accused of cheating by using a system of finger signals that could indicate the number of hearts held by the players. Reese and Schapiro denied the accusation, and eventually a hearing was held by the British bridge league. The hearing was in the form of a legal proceeding with prosecution and defense teams, both having the power to call and cross-examine witnesses. During the course of the proceeding, the prosecutor examined specific hands played by Reese and Schapiro and claimed that their playing these hands was consistent with the hypothesis that they were guilty of having illicit knowledge of the heart suit. At this point, the defense attorney pointed out that their play of these hands was also perfectly consistent with their standard line of play. However, the prosecution then argued that as long as their play was consistent with the hypothesis of guilt, it must be counted as evidence toward that hypothesis. What do you think of the reasoning of the prosecution?

Solution

The problem is basically one of determining how the introduction of new evidence (in this example, the playing of the hands) affects the probability of a particular hypothesis. If we let H denote a particular hypothesis (such as the hypothesis that Reese and Schapiro are guilty) and E the new evidence, then

$$P(H|E)=P(HE)P(E)=P(E|H)P(H)P(E|H)P(H)+P(E|Hc)[1-P(H)] \tag{3.2}$$

where *P(H)* is our evaluation of the likelihood of the hypothesis before the introduction of the new evidence. The new evidence will be in support of the hypothesis whenever it makes the hypothesis more likely—that is, whenever $P(H|E)\geq P(H)$. From **Equation (3.2)**, this will be the case whenever

$$P(E|H)\geq P(E|H)P(H)+P(E|Hc)[1-P(H)]$$

or, equivalently, whenever

$$P(E|H)\geq P(E|Hc)$$

In other words, any new evidence can be considered to be in support of a particular hypothesis only if its occurrence is more likely when the hypothesis is true than when it is false. In fact, the new probability of the hypothesis depends on its initial probability and the ratio of these conditional probabilities, since, from **Equation (3.2)**,

$$P(H|E)=P(H)P(H)+[1-P(H)]P(E|Hc)P(E|H)$$

Hence, in the problem under consideration, the play of the cards can be considered to support the hypothesis of guilt only if such play would have been more likely if the partnership were cheating than if it were not. As the prosecutor never made this claim, his assertion that the evidence is in support of the guilt hypothesis is invalid.

Example 3h

Twins can be either identical or fraternal. Identical, also called monozygotic, twins form when a single fertilized egg splits into two genetically identical parts. Consequently, identical twins always have the same set of genes. Fraternal, also called dizygotic, twins develop when two eggs are fertilized and implant in the uterus. The genetic connection of fraternal twins is no more or less the same as siblings born at separate times. A Los Angeles County, California, scientist wishing to know the current fraction of twin pairs born in the county that are identical twins has assigned a county statistician to study this issue. The statistician initially requested each hospital in the county to record all twin births, indicating whether or not the resulting twins were identical. The hospitals, however, told her that to determine whether newborn twins were identical was not a simple task, as it involved the permission of the twins' parents to perform complicated and expensive DNA studies that the hospitals could not afford. After some deliberation, the statistician just asked the hospitals for data listing all twin births along with an indication as to whether the twins were of the same sex. When such data indicated that approximately 64 percent of twin births were same-sexed, the statistician declared that approximately 28 percent of all twins were identical. How did she come to this conclusion?

Solution

The statistician reasoned that identical twins are always of the same sex, whereas fraternal twins, having the same relationship to each other as any pair of siblings, will have probability 1/2 of being of the same sex. Letting I be the event that a pair of twins is identical, and SS be the event that a pair of twins is of the same sex, she computed the probability P(SS) by conditioning on whether the twin pair was identical. This gave

$$P(SS)=P(SS|I)P(I)+P(SS|Ic)P(Ic)$$

or

$$P(SS)=1\times P(I)+12\times[1-P(I)]=12+12\,P(I)$$

which, using that $P(SS)\approx.64$ yielded the result

$$P(I)\approx.28$$

The change in the probability of a hypothesis when new evidence is introduced can be expressed compactly in terms of the change in the *odds* of that hypothesis, where the concept of odds is defined as follows.

> Definition
>
> The odds of an event A are defined by
>
> $$P(A)P(Ac)=P(A)1-P(A)$$
>
> That is, the odds of an event A tell how much more likely it is that the event A occurs than it is that it does not occur. For instance, if $P(A)=23$, then $P(A)=2P(Ac)$, so the odds are 2. If the odds are equal to α, then it is common to say that the odds are "α to 1" in favor of the hypothesis.

Consider now a hypothesis H that is true with probability P(H), and suppose that new evidence E is introduced. Then, the conditional probabilities, given the evidence E, that H is true and that H is not true are respectively given by

$$P(H|E)=P(E|H)P(H)P(E)\quad P(Hc|E)=P(E|Hc)P(Hc)P(E)$$

Therefore, the new odds after the evidence E has been introduced are

$$P(H|E)P(Hc|E)=P(H)P(Hc)P(E|H)P(E|Hc) \tag{3.3}$$

That is, the new value of the odds of H is the old value multiplied by the ratio of the conditional probability of the new evidence given that H is true to the conditional probability given that H is not true. Thus, **Equation (3.3)** verifies the result of **Example 3f**, since the odds, and thus the probability of H, increase whenever the new evidence is more likely when H is true than when it is false. Similarly, the odds decrease whenever the new evidence is more likely when H is false than when it is true.

Example 3i

An urn contains two type A coins and one type B coin. When a type A coin is flipped, it comes up heads with probability 1/4, whereas when a type B coin is flipped, it comes up heads with probability 3/4. A coin is randomly chosen from the urn and flipped. Given that the flip landed on heads, what is the probability that it was a type A coin?

Solution

Let A be the event that a type A coin was flipped, and let B=Ac be the event that a type B coin was flipped. We want P(A| heads), where heads is the event that the flip landed on heads. From **Equation (3.3)**, we see that

$$P(A|heads)P(Ac|heads)=P(A)P(B)P(\ heads|A)P(\ heads|B)=2/31/3\ 1/43/4=2/3$$

Hence, the odds are 2/3:1, or, equivalently, the probability is 2/5 that a type A coin was flipped.

Equation (3.1) may be generalized as follows: Suppose that F1,F2,…,Fn are mutually exclusive events such that

$$\cup i=1nFi=S$$

In other words, exactly one of the events F1,F2,…,Fn must occur. By writing

$$E=\cup i=1nEFi$$

and using the fact that the events EFi,i=1,…,n are mutually exclusive, we obtain

$$P(E)=\sum i=1nP(EFi)=\sum i=1nP(E|Fi)P(Fi) \tag{3.4}$$

Thus, **Equation (3.4)**, often referred to as the *law of total probability*, shows how, for given events F1,F2,…,Fn, of which one and only one must occur, we can compute P(E) by first conditioning on which one of the Fi occurs. That is, **Equation (3.4)** states that P(E) is equal to a weighted average of P(E|Fi), each term being weighted by the probability of the event on which it is conditioned.

Example 3j

In **Example 5j** of **Chapter 2**, we considered the probability that, for a randomly shuffled deck, the card following the first ace is some specified card, and we gave a combinatorial argument to show that this probability is 152. Here is a probabilistic argument based on conditioning: Let E be the event that the card following the first ace is some specified card, say, card x. To compute P(E), we ignore card x and condition on the relative ordering of the other *51* cards in the deck. Letting O be the ordering gives

$$P(E)=\sum OP(E|O)P(O)$$

Now, given O, there are *52* possible orderings of the cards, corresponding to having card x being the ith card in the deck, i=1,…,52. But because all 52! possible orderings were initially equally likely, it follows that, conditional on O, each of the *52* remaining possible orderings is equally likely. Because card x will follow the first ace for only one of these orderings, we have $P(E|O)=1/52$, implying that $P(E)=1/52$.

Again, let F1,…,Fn be a set of mutually exclusive and exhaustive events (meaning that exactly one of these events must occur).

Suppose now that E has occurred and we are interested in determining which one of the Fj also occurred. Then, by **Equation (3.4)**, we have the following proposition.

Proposition 3.1

$$P(Fj|E)=P(EFj)P(E)=P(E|Fj)P(Fj)\sum i=1nP(E|Fi)P(Fi) \tag{3.5}$$

Equation (3.5) is known as *Bayes's formula*, after the English philosopher Thomas Bayes. If we think of the events Fj as being possible "hypotheses" about some subject matter, then Bayes's formula may be interpreted as showing us how opinions about these hypotheses held before the experiment was carried out [that is, the P(Fj)] should be modified by the evidence produced by the experiment.

Example 3k

A plane is missing, and it is presumed that it was equally likely to have gone down in any of 3 possible regions. Let $1-\beta i, i=1,2,3$, denote the probability that the plane will be found upon a search of the ith region when the plane is, in fact, in that region. (The constants βi are called *overlook probabilities*, because they represent the probability of overlooking the plane; they are generally attributable to the geographical and environmental conditions of the regions.) What is the conditional probability that the plane is in the ith region given that a search of region 1 is unsuccessful?

Solution

Let Ri,i=1,2,3, be the event that the plane is in region i, and let E be the event that a search of region 1 is unsuccessful. From Bayes's formula, we obtain

$$P(R1|E)=P(ER1)P(E)=P(E|R1)P(R1)\sum i=13P(E|Ri)P(Ri)=(\beta 1)13(\beta 1)13+(1)13+(1)13=\beta 1\beta 1+2$$

For j=2, 3,

$$P(Rj|E)=P(E|Rj)P(Rj)P(E)=(1)13(\beta 1)13+13+13=1\beta 1+2 \quad j=2,3$$

Note that the updated (that is, the conditional) probability that the plane is in region j, given the information that a search of region 1 did not find it, is greater than the initial probability that it was in region j when j≠1 and is less than the initial probability when j=1. This statement is certainly intuitive, since not finding the plane in region 1 would seem to decrease its chance of being in that region and increase its chance of being elsewhere. Further, the conditional probability that the plane is in region 1 given an unsuccessful search of that region is an increasing function of the overlook probability β1. This statement is also intuitive, since the larger β1 is, the more it is reasonable to attribute the unsuccessful search to “bad luck” as opposed to the plane’s not being there. Similarly, P(Rj|E),j≠1, is a decreasing function of β1.

The next example has often been used by unscrupulous probability students to win money from their less enlightened friends.

Example 3l

Suppose that we have 3 cards that are identical in form, except that both sides of the first card are colored red, both sides of the second card are colored black, and one side of the third card is colored red and the other side black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground. If the upper side of the chosen card is colored red, what is the probability that the other side is colored black?

Solution

Let *RR, BB*, and *RB* denote, respectively, the events that the chosen card is all red, all black, or the red–black card. Also, let R be the event that the upturned side of the chosen card is red. Then, the desired probability is obtained by

P(RB|R)=P(RB∩R)P(R)=P(R|RB)P(RB)P(R|RR)P(RR)+P(R|RB)P(RB)+P(R|BB)P(BB)=(12)(13)(1)(13)+(12)(13)+0(13)=13

Hence, the answer is 13. Some students guess 12 as the answer by incorrectly reasoning that given that a red side appears, there are two equally likely possibilities: that the card is the all-red card or the red–black card. Their mistake, however, is in assuming that these two possibilities are equally likely. For, if we think of each card as consisting of two distinct sides, then we see that there are 6 equally likely outcomes of the experiment—namely, R1,R2,B1,B2,R3,B3—where the outcome is R1 if the first side of the all-red card is turned face up, R2 if the second side of the all-red card is turned face up, R3 if the red side of the red–black card is turned face up, and so on. Since the other side of the upturned red side will be black only if the outcome is R3, we see that the desired probability is the conditional probability of R3 given that either R1 or R2 or R3 occurred, which obviously equals 13.

Example 3m

A new couple, known to have two children, has just moved into town. Suppose that the mother is encountered walking with one of her children. If this child is a girl, what is the probability that both children are girls?

Solution

Let us start by defining the following events:

G_1: The first (that is, the oldest) child is a girl.

G_2: The second child is a girl.

G: The child seen with the mother is a girl.

Also, let B1,B2, and B denote similar events, except that “girl” is replaced by “boy.” Now, the desired probability is P(G1G2|G), which can be expressed as follows:

P(G1G2|G)=P(G1G2G)P(G)=P(G1G2)P(G)

Also,

P(G)=P(G|G1G2)P(G1G2)+P(G|G1B2)P(G1B2)+P(G|B1G2)P(B1G2)+P(G|B1B2)P(B1B2)=P(G1G2)+P(G|G1B2)P(G1B2)+P(G|B1G2)P(B1G2)

where the final equation used the results P(G|G1G2)=1 and P(G|B1B2)=0. If we now make the usual assumption that all 4 gender possibilities are equally likely, then we see that

P(G1G2|G)=1414+P(G|G1B2)/4+P(G|B1G2)/4=11+P(G|G1B2)+P(G|B1G2)

Thus, the answer depends on whatever assumptions we want to make about the conditional probabilities that the child seen with the mother is a girl given the event G1B2 and that the child seen with the mother is a girl given the event G2B1. For instance, if we want to assume, on the one hand, that, independently of the genders of the children, the child walking with the mother is the elder child with some probability p, then it would follow that

P(G|G1B2)=p=1−P(G|B1G2)

implying under this scenario that

P(G1G2|G)=12

If, on the other hand, we were to assume that if the children are of different genders, then the mother would choose to walk with the girl with probability q, independently of the birth order of the children, then we would have

$$P(G|G1B2)=P(G|B1G2)=q$$

implying that

$$P(G1G2|G)=11+2q$$

For instance, if we took q=1, meaning that the mother would always choose to walk with a daughter, then the conditional probability that she has two daughters would be 13, which is in accord with **Example 2b** because seeing the mother with a daughter is now equivalent to the event that she has at least one daughter.

Hence, as stated, the problem is incapable of solution. Indeed, even when the usual assumption about equally likely gender probabilities is made, we still need to make additional assumptions before a solution can be given. This is because the sample space of the experiment consists of vectors of the form s1,s2, i, where s1 is the gender of the older child, s2 is the gender of the younger child, and i identifies the birth order of the child seen with the mother. As a result, to specify the probabilities of the events of the sample space, it is not enough to make assumptions only about the genders of the children; it is also necessary to assume something about the conditional probabilities as to which child is with the mother given the genders of the children.

Example 3n

A bin contains 3 types of disposable flashlights. The probability that a type 1 flashlight will give more than 100 hours of use is .7, with the corresponding probabilities for type 2 and type 3 flashlights being .4 and .3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and 50 percent are type 3.

a. What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
b. Given that a flashlight lasted more than 100 hours, what is the conditional probability that it was a type j flashlight, j=1,2,3

Solution

a. Let A denote the event that the flashlight chosen will give more than 100 hours of use, and let Fj be the event that a type j flashlight is chosen, j=1,2,3. To compute P(A), we condition on the type of the flashlight, to obtain
$P(A)=P(A|F1)P(F1)+P(A|F2)P(F2)+P(A|F3)P(F3)=(.7)(.2)+(.4)(.3)+(.3)(.5)=.41$

   There is a 41 percent chance that the flashlight will last for more than 100 hours.
b. The probability is obtained by using Bayes's formula:
$P(Fj|A)=P(AFj)P(A)=P(A|Fj)P(Fj).41$

   Thus,
$P(F1|A)=(.7)(.2)/.41=14/41P(F2|A)=(.4)(.3)/.41=12/41P(F3|A)=(.3)(.5)/.41=15/41$

   For instance, whereas the initial probability that a type 1 flashlight is chosen is only .2, the information that the flashlight has lasted more than 100 hours raises the probability of this event to $14/41\approx.341$.

Example 3o

A crime has been committed by a solitary individual, who left some DNA at the scene of the crime. Forensic scientists who studied the recovered DNA noted that only five strands could be identified and that each innocent person, independently, would have a probability of 10^{-5} of having his or her DNA match on all five strands. The district attorney supposes that the perpetrator of the crime could be any of the 1 million residents of the town. Ten thousand of these residents have been released from prison within the past 10 years; consequently, a sample of their DNA is on file. Before any checking of the DNA file, the district attorney thinks that each of the 10,000 ex-criminals has probability α of being guilty of the new crime, whereas each of the remaining 990,000 residents has probability β, where $\alpha=c\beta$. (That is, the district attorney supposes that each recently released convict is c times as likely to be the crime's perpetrator as is each town member who is not a recently released convict.) When the DNA that is analyzed is compared against the database of the 10,000 ex-convicts, it turns out that A. J. Jones is the only one whose DNA matches the profile. Assuming that the district attorney's estimate of the relationship between α and β is accurate, what is the probability that A. J. is guilty?

Solution

To begin, note that because probabilities must sum to 1, we have

$$1=10{,}000\alpha+990{,}000\beta=(10{,}000c+990{,}000)\beta$$

Thus,

$$\beta=110{,}000c+990{,}000, \quad \alpha=c10{,}000c+990{,}000$$

Now, let G be the event that A. J. is guilty, and let M denote the event that A. J. is the only one of the 10,000 on file to have a match. Then,

$$P(G|M)=P(GM)P(M)=P(G)P(M|G)P(M|G)P(G)+P(M|Gc)P(Gc)$$

On the one hand, if A. J. is guilty, then he will be the only one to have a DNA match if none of the others on file have a match. Therefore,

$$P(M|G)=(1-10-5)9999$$

On the other hand, if A. J. is innocent, then in order for him to be the only match, his DNA must match (which will occur with probability 10−5), all others in the database must be innocent, and none of these others can have a match. Now, given that A. J. is innocent, the conditional probability that all the others in the database are also innocent is

$$P(\text{ all others innocent}|AJ \text{ innocent})=P(\text{ all in database innocent})P(AJ \text{ innocent})=1-10,000\alpha 1-\alpha$$

Also, the conditional probability, given their innocence, that none of the others in the database will have a match is (1−10−5)9999. Therefore,

$$P(M|Gc)=10-5(1-10,000\alpha 1-\alpha)(1-10-5)9999$$

Because P(G)=α, the preceding formula gives

$$P(G|M)=\alpha\alpha+10-5(1-10,000\alpha)=1.9+10-5\alpha$$

Thus, if the district attorney's initial thoughts were that an arbitrary ex-convict was 100 times more likely to have committed the crime than was a nonconvict (that is, c=100), then α=119,900 and

$$P(G|M)=11.099\approx 0.9099$$

If the district attorney initially thought that the appropriate ratio was c=10, then α=1109,000 and

$$P(G|M)=11.99\approx 0.5025$$

If the district attorney initially thought that the criminal was equally likely to be any of the members of the town(c=1), then α=10−6 and

$$P(G|M)=110.9\approx 0.0917$$

Thus, the probability ranges from approximately 9 percent when the district attorney's initial assumption is that all the members of the population have the same chance of being the perpetrator to approximately 91 percent when she assumes that each ex-convict is 100 times more likely to be the criminal than is a specified townsperson who is not an ex-convict.

3.4 Independent Events

The previous examples in this chapter show that P(E|F), the conditional probability of E given F, is not generally equal to P(E), the unconditional probability of E. In other words, knowing that F has occurred generally changes the chances of E's occurrence. In the special cases where P(E|F) does in fact equal P(E), we say that E is independent of F. That is, E is independent of F if knowledge that F has occurred does not change the probability that E occurs.

Since P(E|F)=P(EF)/P(F), it follows that E is independent of F if

$$P(EF)=P(E)P(F) \tag{4.1}$$

The fact that **Equation (4.1)** is symmetric in E and F shows that whenever E is independent of F, F is also independent of E. We thus have the following definition.

> Definition
>
> Two events E and F are said to be *independent* if **Equation (4.1)** holds.
>
> Two events E and F that are not independent are said to be *dependent*.

Example 4a

A card is selected at random from an ordinary deck of 52 playing cards. If E is the event that the selected card is an ace and F is the event that it is a spade, then E and F are independent. This follows because P(EF)=152, whereas P(E)=452 and P(F)=1352.

Example 4b

Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If E is the event that the first coin lands on heads and F the event that the second lands on tails, then E and F are independent, since P(EF)=P({(H,T)})=14, whereas P(E)=P({(H,H),(H,T)})=12 and P(F)=P({(H,T),(T,T)})=12.

Example 4c

Suppose that we toss 2 fair dice. Let E1 denote the event that the sum of the dice is 6 and F denote the event that the first die equals 4. Then

$$P(E1F)=P(\{(4,2)\})=136$$

whereas

$$P(E1)P(F)=(536)(16)=5216$$

Hence, E1 and F are not independent. Intuitively, the reason for this is clear because if we are interested in the possibility of throwing a 6 (with 2 dice), we shall be quite happy if the first die lands on 4 (or, indeed, on any of the numbers 1, 2, 3, 4, and 5), for then we shall still have a possibility of getting a total of 6. If, however, the first die landed on 6, we would be unhappy because we would no longer have a chance of getting a total of 6. In other words, our chance of getting a total of 6 depends on the outcome of the first die; thus, E1 and F cannot be independent.

Now, suppose that we let E2 be the event that the sum of the dice equals 7. Is E2 independent of F? The answer is yes, since

$$P(E2F)=P(\{(4,3)\})=136$$

whereas

$$P(E2)P(F)=(16)(16)=136$$

We leave it for the reader to present the intuitive argument why the event that the sum of the dice equals 7 is independent of the outcome on the first die.

Example 4d

If we let E denote the event that the next president is a Republican and F the event that there will be a major earthquake within the next year, then most people would probably be willing to assume that E and F are independent. However, there would probably be some controversy over whether it is reasonable to assume that E is independent of G, where G is the event that there will be a recession within two years after the election.

We now show that if E is independent of F, then E is also independent of Fc.

Proposition 4.1

If E and F are independent, then so are E and Fc.

Proof. Assume that E and F are independent. Since E=EF∪EFc and *EF* and EFc are obviously mutually exclusive, we have

$$P(E)=P(EF)+P(EFc)=P(E)P(F)+P(EFc)$$

or, equivalently,

$$P(EFc)=P(E)[1-P(F)]=P(E)P(Fc)$$

and the result is proved.

Thus, if E is independent of F, then the probability of E's occurrence is unchanged by information as to whether or not F has occurred.

Suppose now that E is independent of F and is also independent of G. Is E then necessarily independent of *FG*? The answer, somewhat surprisingly, is no, as the following example demonstrates.

Example 4e

Two fair dice are thrown. Let E denote the event that the sum of the dice is 7. Let F denote the event that the first die equals 4 and G denote the event that the second die equals 3. From **Example 4c**, we know that E is independent of F, and the same reasoning as applied there shows that E is also independent of G; but clearly, E is not independent of *FG* [since P(E|FG)=1].

It would appear to follow from **Example 4e** that an appropriate definition of the independence of three events E, F, and G would have to go further than merely assuming that all of the (32) pairs of events are independent. We are thus led to the following definition.

Definition

Three events E, F, and G are said to be independent if

$$P(EFG)=P(E)P(F)P(G)P(EF)=P(E)P(F)P(EG)=P(E)P(G)P(FG)=P(F)P(G)$$

Note that if E, F, and G are independent, then E will be independent of any event formed from F and G. For instance, E is independent of F∪G, since

$$P[E(F\cup G)]=P(EF\cup EG)=P(EF)+P(EG)-P(EFG)=P(E)P(F)+P(E)P(G)-P(E)P(FG)=P(E)[P(F)+P(G)-P(FG)]=P(E)P(F\cup G)$$

Of course, we may also extend the definition of independence to more than three events. The events E1,E2,…,En are said to be independent if for every subset E1′,E2′,…,Er′,r≤n of these events,

$$P(E1'E2'\cdots Er')=P(E1')P(E2')\cdots P(Er')$$

Finally, we define an infinite set of events to be independent if every finite subset of those events is independent.

Sometimes, a probability experiment under consideration consists of performing a sequence of subexperiments. For instance, if the experiment consists of continually tossing a coin, we may think of each toss as being a subexperiment. In many cases, it is reasonable to assume that the outcomes of any group of the subexperiments have no effect on the probabilities of the outcomes of the other subexperiments. If such is the case, we say that the subexperiments are independent. More formally, we say that the subexperiments are independent if E1,E2,…,En,… is necessarily an independent sequence of events whenever Ei is an event whose occurrence is completely determined by the outcome of the ith subexperiment.

If each subexperiment has the same set of possible outcomes, then the subexperiments are often called *trials*.

Example 4f

An infinite sequence of independent trials is to be performed. Each trial results in a success with probability p and a failure with probability 1−p. What is the probability that

a. at least 1 success occurs in the first n trials;
b. exactly k successes occur in the first n trials;
c. all trials result in successes?

Solution

In order to determine the probability of at least 1 success in the first n trials, it is easiest to compute first the probability of the complementary event: that of no successes in the first n trials. If we let Ei denote the event of a failure on the ith trial, then the probability of no successes is, by independence,

$$P(E1E2\cdots En)=P(E1)P(E2)\cdots P(En)=(1-p)n$$

Hence, the answer to part (a) is 1−(1−p)n.

To compute the answer to part (b), consider any particular sequence of the first n outcomes containing k successes and n−k failures. Each one of these sequences will, by the assumed independence of trials, occur with probability pk(1−p)n−k. Since there are (nk) such sequences [there are n!/k!(n−k)! permutations of k successes and n−k failures], the desired probability in part (b) is

$$P\{\text{exactly k successes}\}=(nk)pk(1-p)n-k$$

To answer part (c), we note that, by part (a), the probability of the first n trials all resulting in success is given by

$$P(E1cE2c\cdots Enc)=pn$$

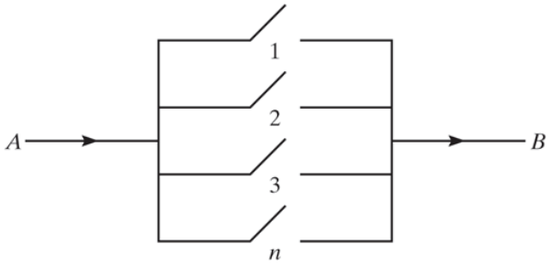
Thus, using the continuity property of probabilities (**Section 2.6**), we see that the desired probability is given by

$$P(\cap i=1\infty Eic)=P(limn\to\infty\cap i=1\infty Eic)=limn\to\infty P(\cap i=1\infty Eic)=limnpn=\{\ 0if\ p<11if\ p=1$$

Example 4g

A system composed of n separate components is said to be a parallel system if it functions when at least one of the components functions. (See **Figure 3.2**.) For such a system, if component i, which is independent of the other components, functions with probability pi,i=1,…,n, what is the probability that the system functions?

Figure 3.2 Parallel system: functions if current flows from *A* to *B*.



Solution

Let Ai denote the event that component i functions. Then,

P{ system functions }=1−P{ system does not function }=1−P{ all components do not function }=1−P(∩iAic)=1−∏i=1i(1−pi) by independence

Example 4h

Independent trials consisting of rolling a pair of fair dice are performed. What is the probability that an outcome of 5 appears before an outcome of 7 when the outcome of a roll is the sum of the dice?

Solution

If we let En denote the event that no 5 or 7 appears on the first n−1 trials and a 5 appears on the nth trial, then the desired probability is

P(∪n=1∞En)=∑n=1∞P(En)

Now, since P{5 on any trial}=436 and P{7 on any trial}=636, we obtain, by the independence of trials,

P(En)=(1−1036)n−1436

Thus,

P(∪n=1∞En)=19∑n=1∞(1318)n−1=1911−1318=25

This result could also have been obtained by the use of conditional probabilities. If we let E be the event that a 5 occurs before a 7, then we can obtain the desired probability, P(E), by conditioning on the outcome of the first trial, as follows: Let F be the event that the first trial results in a 5, let G be the event that it results in a 7, and let H be the event that the first trial results in neither a 5 nor a 7. Then, conditioning on which one of these events occurs gives

P(E)=P(E|F)P(F)+P(E|G)P(G)+P(E|H)P(H)

However,

P(E|F)=1P(E|G)=0P(E|H)=P(E)

The first two equalities are obvious. The third follows because if the first outcome results in neither a 5 nor a 7, then at that point the situation is exactly as it was when the problem first started—namely, the experimenter will continually roll a pair of fair dice until either a 5 or 7 appears. Furthermore, the trials are independent; therefore, the outcome of the first trial will have no effect on subsequent rolls of the dice. Since P(F)=436,P(G)=636, and P(H)=2636, it follows that

P(E)=19+P(E)1318

or

P(E)=25

The reader should note that the answer is quite intuitive. That is, because a 5 occurs on any roll with probability 436 and a 7 with probability 636, it seems intuitive that the odds that a 5 appears before a 7 should be 6 to 4 against. The probability should then be 410, as indeed it is.

The same argument shows that if E and F are mutually exclusive events of an experiment, then, when independent trials of the experiment are performed, the event E will occur before the event F with probability

P(E)P(E)+P(F)

Example 4i

Suppose there are n types of coupons and that each new coupon collected is, independent of previous selections, a type i coupon with probability pi, ∑i=1npi=1. Suppose k coupons are to be collected. If Ai is the event that there is at least one type i coupon among those collected, then, for i≠j, find

a. P(Ai)
b. P(Ai∪Aj)
c. P(Ai|Aj)

Solution

$P(A_i)=1-P(A_i^c)=1-P\{\text{no coupon is type } i\}=1-(1-p_i)^k$

where the preceding used that each coupon is, independently, not of type i with probability $1-p_i$. Similarly,

$P(A_i \cup A_j)=1-P((A_i \cup A_j)^c)=1-P\{\text{no coupon is either type } i \text{ or type } j\}=1-(1-p_i-p_j)^k$

where the preceding used that each coupon is, independently, neither of type i nor type j with probability $1-p_i-p_j$.

To determine $P(A_i|A_j)$, we will use the identity

$P(A_i \cup A_j)=P(A_i)+P(A_j)-P(A_iA_j)$

which, in conjunction with parts (a) and (b), yields

$P(A_iA_j)=1-(1-p_i)^k+1-(1-p_j)^k-[1-(1-p_i-p_j)^k]=1-(1-p_i)^k-(1-p_j)^k+(1-p_i-p_j)^k$

Consequently,

$P(A_i|A_j)=\frac{P(A_iA_j)}{P(A_j)}=\frac{1-(1-p_i)^k-(1-p_j)^k+(1-p_i-p_j)^k}{1-(1-p_j)^k}$

The next example presents a problem that occupies an honored place in the history of probability theory. This is the famous *problem of the points*. In general terms, the problem is this: Two players put up stakes and play some game, with the stakes to go to the winner of the game. An interruption requires them to stop before either has won and when each has some sort of a "partial score." How should the stakes be divided?

This problem was posed to the French mathematician Blaise Pascal in 1654 by the Chevalier de Méré, who was a professional gambler at that time. In attacking the problem, Pascal introduced the important idea that the proportion of the prize deserved by the competitors should depend on their respective probabilities of winning if the game were to be continued at that point. Pascal worked out some special cases and, more importantly, initiated a correspondence with the famous Frenchman Pierre de Fermat, who had a reputation as a great mathematician. The resulting exchange of letters not only led to a complete solution to the problem of the points, but also laid the framework for the solution to many other problems connected with games of chance. This celebrated correspondence, considered by some as the birth date of probability theory, was also important in stimulating interest in probability among the mathematicians in Europe, for Pascal and Fermat were both recognized as being among the foremost mathematicians of the time. For instance, within a short time of their correspondence, the young Dutch mathematician Christiaan Huygens came to Paris to discuss these Problems and solutions, and interest and activity in this new field grew rapidly.

Example 4j The problem of the points

Independent trials resulting in a success with probability p and a failure with probability $1-p$ are performed. What is the probability that n successes occur before m failures? If we think of A and B as playing a game such that A gains 1 point when a success occurs and B gains 1 point when a failure occurs, then the desired probability is the probability that A would win if the game were to be continued in a position where A needed n and B needed m more points to win.

Solution

We shall present two solutions. The first comes from Pascal and the second from Fermat.

Let us denote by $P_{n,m}$ the probability that n successes occur before m failures. By conditioning on the outcome of the first trial, we obtain

$P_{n,m}=pP_{n-1,m}+(1-p)P_{n,m-1} n \geq 1, m \geq 1$

(Why? Reason it out.) Using the obvious boundary conditions $P_{n,0}=0, P_{0,m}=1$, we can solve these equations for $P_{n,m}$. Rather than go through the tedious details, let us instead consider Fermat's solution.

Fermat argued that in order for n successes to occur before m failures, it is necessary and sufficient that there be at least n successes in the first $m+n-1$ trials. (Even if the game were to end before a total of $m+n-1$ trials were completed, we could still imagine that the necessary additional trials were performed.) This is true, for if there are at least n successes in the first $m+n-1$ trials, there could be at most $m-1$ failures in those $m+n-1$ trials; thus, n successes would occur before m failures. If, however, there were fewer than n successes in the first $m+n-1$ trials, there would have to be at least m failures in that same number of trials; thus, n successes would not occur before m failures.

Hence, since, as shown in **Example 4f**, the probability of exactly k successes in $m+n-1$ trials is $\binom{m+n-1}{k}p^k(1-p)^{m+n-1-k}$, it follows that the desired probability of n successes before m failures is

$P_{n,m}=\sum_{k=n}^{m+n-1}\binom{m+n-1}{k}p^k(1-p)^{m+n-1-k}$

The following example gives another instance where determining the probability that a player wins a match is made easier by assuming that the play continues even after the match winner has been determined.

Example 4k Service protocol in a serve and rally game

Consider a serve and rally match (such as volleyball, badminton, or squash) between two players, A and B. The match consists of a sequence of rallies, with each rally beginning with a serve by one of the players and continuing until one of the players has won the rally. The winner of the rally receives a point, and the match ends when one of the players has won a total of n points, with that player being declared the winner of the match. Suppose whenever a rally begins with A as the server, that A wins that rally with probability p_A and that B wins it with probability $q_A=1-p_A$, and that a rally that begins with B as the server is won by A with probability p_B and by B with probability $q_B=1-p_B$. Player A is to be the initial server. There are two possible server protocols that are under consideration: "winner serves," which means that the winner of a rally is the server for the next rally, or "alternating serve," which means that the server alternates from rally to rally, so that no two consecutive rallies have the same server. Thus, for instance, if n=3, then the successive servers under the "winner serves" protocol would be A,A,B,A,A if A wins the first point, then B the next, then A wins the next two. On the other hand, the sequence of servers under the "alternating serve" protocol will always be A,B,A,B,A,... until the match winner is decided. If you were player A, which protocol would you prefer?

Solution

Surprisingly, it turns out that it makes no difference, in that the probability that A is the match winner is the same under either protocol. To show that this is the case, it is advantageous to suppose that the players continue to play until a

total of 2n−1 rallies have been completed. The first player to win n rallies would then be the one who has won at least n of the 2n−1 rallies. To begin, note that if the alternating serve protocol is being used, then player A will serve exactly n times and player B will serve exactly n−1 times in the 2n−1 rallies.

Now consider the winner serve protocol, again assuming that the players continue to play until 2n−1 rallies have been completed. Because it makes no difference who serves the “extra rallies” after the match winner has been decided, suppose that at the point at which the match has been decided (because one of the players has won n points), the remainder (if there are any) of the 2n−1 rallies are all served by the player who lost the match. Note that this modified service protocol does not change the fact that the winner of the match will still be the player who wins at least n of the 2n−1 rallies. We claim that under this modified service protocol, A will always serve n times and B will always serve n−1 times. Two cases show this.

Case 1: A wins the match.

Because A serves first, it follows that A’s second serve will immediately follow A’s first point; A’s third serve will immediately follow A’s second point; and, in particular, A’s nth serve will immediately follow A’s (n−1) point. But this will be the last serve of A before the match result is decided. This is so because either A will win the point on that serve and so have n points, or A will lose the point and so the serve will switch to, and remain with, B until A wins point number n. Thus, provided that A wins the match, it follows that A would have served a total of n times at the moment the match is decided. Because, by the modified service protocol, A will never again serve, it follows in this case that A serves exactly n times.

Case 2: B wins the match.

Because A serves first, B’s first serve will come immediately after B’s first point; B’s second serve will come immediately after B’s second point; and, in particular, B’s (n−1) serve will come immediately after B’s (n−1) point. But that will be the last serve of B before the match is decided because either B will win the point on that serve and so have n points, or B will lose the point and so the serve will switch to, and remain with, A until B wins point number n. Thus, provided that B wins the match, we see that B would have served a total of n−1 times at the moment the match is decided. Because, by the modified service protocol, B will never again serve, it follows in this case that B serves exactly n−1 times, and, as there are a total of 2n−1 rallies, that A serves exactly n times.

Thus, we see that under either protocol, A will always serve n times and B will serve n−1 times and the winner of the match will be the one who wins at least n points. But since A wins each rally that he serves with probability pA and wins each rally that B serves with probability pB it follows that the probability that A is the match winner is, under either protocol, equal to the probability that there are at least n successes in 2n−1 independent trials, when n of these trials result in a success with probability pA and the other n−1 trials result in a success with probability pB. Consequently, the win probabilities for both protocols are the same.

Our next two examples deal with gambling problems, with the first having a surprisingly elegant analysis.*

*The remainder of this section should be considered optional.

Example 4l

Suppose that initially there are r players, with player i having ni units, ni>0,i=1,…,r. At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all n≡∑i=1rni units, with that player designated as the victor. Assuming that the results of successive games are independent and that each game is equally likely to be won by either of its two players, find Pi, the probability that player i is the victor.

Solution

To begin, suppose that there are n players, with each player initially having 1 unit. Consider player i. Each stage she plays will be equally likely to result in her either winning or losing 1 unit, with the results from each stage being independent.

In addition, she will continue to play stages until her fortune becomes either 0 or n. Because this is the same for all n players, it follows that each player has the same chance of being the victor, implying that each player has probability 1/n of being the victor. Now, suppose these n players are divided into r teams, with team i containing ni players, i=1,…,r. Then, the probability that the victor is a member of team i is ni/n. But because

a. team i initially has a total fortune of ni units, i=1,…,r, and
b. each game played by members of different teams is equally likely to be won by either player and results in the fortune of members of the winning team increasing by 1 and the fortune of the members of the losing team decreasing by *1*,

it is easy to see that the probability that the victor is from team i is exactly the probability we desire. Thus, Pi=ni/n. Interestingly, our argument shows that this result does *not* depend on how the players in each stage are chosen.

In the *gambler’s ruin problem*, there are only *2* gamblers, but they are not assumed to be of equal skill.

Example 4m The gambler’s ruin problem

Two gamblers, A and B, bet on the outcomes of successive flips of a coin. On each flip, if the coin comes up heads, A collects 1 unit from B, whereas if it comes up tails, A pays 1 unit to B. They continue to do this until one of them runs out of money. If it is assumed that the successive flips of the coin are independent and each flip results in a head with probability p, what is the probability that A ends up with all the money if he starts with i units and B starts with N−i units?

Solution

Let E denote the event that A ends up with all the money when he starts with i and B starts with N−i, and to make clear the dependence on the initial fortune of A, let Pi=P(E). We shall obtain an expression for *P(E)* by conditioning on the outcome of the first flip as follows: Let H denote the event that the first flip lands on heads; then

Pi=P(E)=P(E|H)P(H)+P(E|Hc)P(Hc)=pP(E|H)+(1−p)P(E|Hc)

Now, given that the first flip lands on heads, the situation after the first bet is that A has i+1 units and B has N−(i+1). Since the successive flips are assumed to be independent with a common probability p of heads, it follows that from that point on, A’s probability of winning all the money is exactly the same as if the game were just starting with A having an initial fortune of i+1 and B having an initial fortune of N−(i+1). Therefore,

P(E|H)=Pi+1

and similarly,

$$P(E|Hc)=Pi-1$$

Hence, letting q=1−p, we obtain

$$Pi=pPi+1+qPi-1i=1,2,\ldots,N-1 \tag{4.2}$$

By making use of the obvious boundary conditions P0=0 and PN=1, we shall now solve **Equation (4.2)**. Since p+q=1, these equations are equivalent to

$$pPi+qPi=pPi+1+qPi-1$$

or

$$Pi+1-Pi=qp(Pi-Pi-1)i=1,2,\ldots,N-1 \tag{4.3}$$

Hence, since P0=0, we obtain, from **Equation (4.3)**,

$$P2-P1=qp(P1-P0)=qpP1P3-P2=qp(P2-P1)=(qp)2P1...Pi-Pi-1=qp(Pi-1-Pi-2)=(qp)i-1P1...PN-PN-1=qp(PN-1-PN-2)=(qp)N-1P1 \tag{4.4}$$

Adding the first i−1 equations of (**4.4**) yields

$$Pi-P1=P1[(qp)+(qp)2+\cdots+(qp)i-1]$$

or

$$Pi=\{1-(q/p)i1-(q/p)P1if\ qp\neq1iP1if\ qp=1$$

Using the fact that PN=1, we obtain

$$P1=\{1-(q/p)1-(q/p)Nif\ p\neq121N \quad if\ p=12$$

Hence,

$$Pi=\{1-(q/p)i1-(q/p)Nif\ p\neq12iNif\ p=12 \tag{4.5}$$

Let Qi denote the probability that B winds up with all the money when A starts with i and B starts with N−i. Then, by symmetry to the situation described, and on replacing p by q and i by N−i, it follows that

$$Qi=\{1-(p/q)N-i1-(p/q)Nif\ q\neq12N-iNif\ q=12$$

Moreover, since q=12 is equivalent to p=12, we have, when q≠12,

$$Pi+Qi=1-(q/p)i1-(q/p)N+1-(p/q)N-i1-(p/q)N=pN-pN(q/p)ipN-qN+qN-qN(p/q)N-iqN-pN=pN-pN-iqi-qN+qipN-ipN-qN=1$$

This result also holds when p=q=12, so

$$Pi+Qi=1$$

In words, this equation states that with probability 1, either A or B will wind up with all of the money; in other words, the probability that the game continues indefinitely with A's fortune always being between 1 and N−1 is zero. (The reader must be careful because, a priori, there are three possible outcomes of this gambling game, not two: Either A wins, or B wins, or the game goes on forever with nobody winning. We have just shown that this last event has probability 0.)

As a numerical illustration of the preceding result, if A were to start with 5 units and B with 10, then the probability of A's winning would be 13 if p were 12, whereas it would jump to

$$1-(23)51-(23)15\approx.87$$

if p were .6.

A special case of the gambler's ruin problem, which is also known as the problem of *duration of play*, was proposed to Huygens by Fermat in 1657. The version Huygens proposed, which he himself solved, was that A and B have 12 coins each. They play for these coins in a game with 3 dice as follows: Whenever 11 is thrown (by either—it makes no difference who rolls the dice), A gives a coin to B. Whenever 14 is thrown, B gives a coin to A. The person who first wins all the coins wins the game. Since P{roll 11}=27216 and P{roll 14}=15216, we see from **Example 4h** that, for A, this is just the gambler's ruin problem with p=1542,i=12, and N=24. The general form of the gambler's ruin problem was solved by the mathematician James Bernoulli and published 8 years after his death in 1713.

For an Application of the gambler's ruin problem to drug testing, suppose that two new drugs have been developed for treating a certain disease. Drug i has a cure rate pi,i=1,2, in the sense that each patient treated with drug i will be cured with probability pi. These cure rates are, however, not known, and we are interested in finding a method for deciding whether p1>p2 or p2>p1. To decide on one of these alternatives, consider the following test: Pairs of patients are to be treated sequentially, with one member of the pair receiving drug 1 and the other drug 2. The results for each pair are determined, and the testing stops when the cumulative number of cures from one of the drugs exceeds the cumulative number of cures from the other by some fixed, predetermined number. More formally, let

$$Xj=\{1if\ the\ patient\ in\ the\ jth\ pair\ that\ receives\ drug\ 1\ is\ cured0otherwiseYj=\{1if\ the\ patient\ in\ the\ jth\ pair\ that\ receives\ drug\ 2\ is\ cured0otherwise$$

For a predetermined positive integer M, the test stops after pair N, where N is the first value of n such that either

$$X1+\cdots+Xn-(Y1+\cdots+Yn)=M$$

or

$$X_1+\cdots+X_n-(Y_1+\cdots+Y_n)=-M$$

In the former case, we assert that $p_1>p_2$ and in the latter that $p_2>p_1$.

In order to help ascertain whether the foregoing is a good test, one thing we would like to know is the probability that it leads to an incorrect decision. That is, for given p_1 and p_2, where $p_1>p_2$, what is the probability that the test will incorrectly assert that $p_2>p_1$? To determine this probability, note that after each pair is checked, the cumulative difference of cures using drug 1 versus drug 2 will go up by 1 with probability $p_1(1-p_2)$—since this is the probability that drug 1 leads to a cure and drug 2 does not—or go down by 1 with probability $(1-p_1)p_2$, or remain the same with probability $p_1p_2+(1-p_1)(1-p_2)$. Hence, if we consider only those pairs in which the cumulative difference changes, then the difference will go up by 1 with probability

$$p=P\{\text{up 1} \mid \text{up 1 or down 1}\}=\frac{p_1(1-p_2)}{p_1(1-p_2)+(1-p_1)p_2}$$

and down by 1 with probability

$$1-p=\frac{p_2(1-p_1)}{p_1(1-p_2)+(1-p_1)p_2}$$

Thus, the probability that the test will assert that $p_2>p_1$ is equal to the probability that a gambler who wins each (one-unit) bet with probability p will go down M before going up M. But **Equation (4.5)**, with $i=M, N=2M$, shows that this probability is given by

$$p=P\{\text{ test asserts that } p_2>p_1 \}=\frac{1-1-(1-pp)^M}{1-(1-pp)^{2M}}=\frac{1-1}{1+(1-pp)^M}=\frac{1}{1+\gamma^M}$$

where

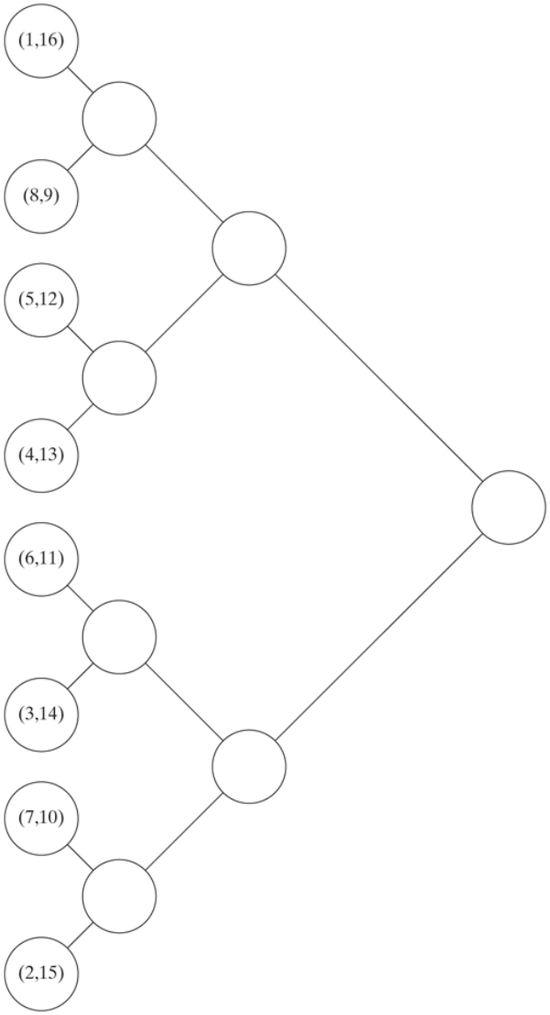
$$\gamma=\frac{1-p}{p}=\frac{p_1(1-p_2)}{p_2(1-p_1)}$$

For instance, if $p_1=.6$ and $p_2=.4$, then the probability of an incorrect decision is .017 when $M=5$ and reduces to .0003 when $M=10$.

Example 4n

A total of 64 teams are selected to play in the end of season NCAA college basketball tournament. These 64 are divided into four groups, called brackets, of size 16 each, with the teams in each bracket being given seedings ranging from 1 (the top rated team in the bracket) to 16 (the lowest rated team in the bracket). The teams in each bracket play each other in a knockout style tournament, meaning a loss knocks a team out of the tournament. Naming a team by its seeding, the schedule of games to be played by the teams in a bracket is as given by the following graph:

Figure 3.3 Ncaa tournament bracket format



Thus, for instance, teams 1 and 16 play a game in round one, as do teams 8 and 9, with the winners of these games then playing each other in round two. Let r(i,j)=r(j,i), i≠j denote the round in which i and j would play if both teams win up to that point. That is, r(i,j)=k if i and j would play in round k if each won its first k−1 games. For instance, r(1,16)=1, r(1,8)=2, r(1,5)=3, r(1,6)=4.

Let us focus on a single one of the brackets, and let us suppose that, no matter what has previously occurred, if i and j ever play each other then i will win with probability pi,j=1−pj,i. Let Pi be the probability that team i is the winner of the bracket, i=1,…,16. Because Pi is the probability that i wins 4 games, we will compute the values P1,…,P16 by determining the quantities Pi(k), i=1,…,16, where Pi(k) is defined to be the probability that i wins its first k games. The probabilities Pi(k) will be determined recursively, first for k=1, then for k=2, then for k=3, and finally for k=4 which will yield Pi=Pi(4).

Let Oi(k)={j:(r(i,j)=k} be the set of possible opponents of i in round k. To compute Pi(k), we will condition on which of the teams in Oi(k) reaches round k. Because a team will reach round k if that team wins its first k−1 games, this gives

$$P_i(k)=\sum_{j\in O_i(k)} P(i \text{ win its first } k \text{ games} \mid j \text{ reaches round } k)\, P_j(k-1) \tag{4.6}$$

Now, because any team that plays a team in Oi(k) in any of rounds 1,…,k−1 is a possible opponent of team i in round k, it follows that all games in rounds 1,…,k−1 involving a team in Oi(k) will be against another team in that set, and thus the results of these games do not affect which teams i would play in its first k−1 games. Consequently, whether team i reaches round k is independent of which team in Oi(k) reaches round k. Hence, for j∈Oi(k)

$$P(i \text{ win its first } k \text{ games} \mid j \text{ reaches round } k)=P(i \text{ win its first } k-1 \text{ games}, i \text{ beats } j \mid j \text{ reaches round } k)=P(i \text{ win its first } k-1 \text{ games})P(i \text{ beats } j \mid i \text{ and } j \text{ reach round } k)=P_i(k-1)\, p_{i,j}$$

where the next to last equality follows because whether i wins its first k−1 games is independent of the event that j wins its first k−1 games. Hence, from (**4.6**) and the preceding equation, we have that

$$P_i(k)=\sum_{j\in O_i(k)} P_i(k-1)\, p_{i,j}\, P_j(k-1)=P_i(k-1)\sum_{j\in O_i(k)} P_j(k-1)\, p_{i,j} \tag{4.7}$$

Starting with Pi(0)=1, the preceding enables us to determine Pi(1) for all i, which then enables us to determine Pi(2) for all i, and so on, up to Pi=Pi(4).

To indicate how to apply the recursive **equations (4.7)**, suppose that pi,j=ji+j. Thus, for instance, the probability that team 2 (the second seed) beats team 7 (the seventh seed) is p2,7=7/9. To compute, Pi=Pi(4), the probability that i wins the bracket, start with the quantities Pi(1), i=1,…,16, equal to the probabilities that each team wins its first game.

$$P1(1)=p1,16=16/17=1-P16(1)P2(1)=p2,15=15/17=1-P15(1)P3(1)=p3,14=14/17=1-P14(1)P4(1)=p4,13=13/17=1-P13(1)P5(1)=p5,12=12/17=1-P12(1)P6(1)=p6,11=11/17=1-P11(1)P7(1)=p7,10=10/17=1-P10(1)P8(1)=p8,9=9/17=1-P9(1)$$

The quantities Pi(2) are then obtained by using the preceding along with the recursion (**4.7**). For instance, because the set of possible opponents of team 1 in round 2 is O1(2)={8,9}, we have that

$$P1(2)=P1(1)(P8(1)p1,8+P9(1)p1,9)=1617(91789+817910)\approx.8415$$

The other quantities Pi(2),…,P16(2) are obtained similarly, and are used to obtain the quantities Pi(3), i=1,…,16, which are then used to obtain Pi=Pi(4), i= 1,…,16.

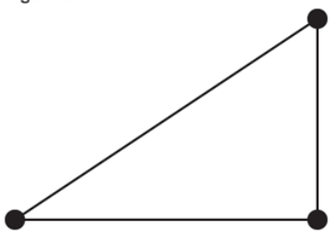
Suppose that we are presented with a set of elements and we want to determine whether at least one member of the set has a certain property. We can attack this question probabilistically by randomly choosing an element of the set in such a way that each element has a positive probability of being selected. Then the original question can be answered by a consideration of the probability that the randomly selected element does not have the property of interest. If this probability is equal to 1, then none of the elements of the set has the property; if it is less than 1, then at least one element of the set has the property.

The final example of this section illustrates this technique.

Example 4o

The complete graph having n vertices is defined to be a set of n points (called vertices) in the plane and the (n2) lines (called edges) connecting each pair of vertices. The complete graph having 3 vertices is shown in **Figure 3.4**. Suppose now that each edge in a complete graph having n vertices is to be colored either red or blue. For a fixed integer k, a question of interest is, Is there a way of coloring the edges so that no set of k vertices has all of its (k2) connecting edges the same color? It can be shown by a probabilistic argument that if n is not too large, then the answer is yes.

Figure 3.4:



The argument runs as follows: Suppose that each edge is, independently, equally likely to be colored either red or blue. That is, each edge is red with probability 12. Number the (nk) sets of k vertices and define the events Ei,i=1,…,(nk) as follows:

Ei={all of the connecting edges of the ith set of k vertices are the same color}

Now, since each of the (k2) connecting edges of a set of k vertices is equally likely to be either red or blue, it follows that the probability that they are all the same color is

$$P(Ei)=2(12)k(k-1)/2$$

Therefore, because

$$P(\cup iEi)\leq\sum iP(Ei) \quad (\text{Boole's inequality})$$

we find that P(∪iEi), the probability that there is a set of k vertices all of whose connecting edges are similarly colored, satisfies

$$P(\cup iEi)\leq(nk)(12)k(k-1)/2-1$$

Hence, if

$$(nk)(12)k(k-1)/2-1<1$$

or, equivalently, if

$$(nk)<2k(k-1)/2-1$$

then the probability that at least one of the (nk) sets of k vertices has all of its connecting edges the same color is less than 1. Consequently, under the preceding condition on n and k, it follows that there is a positive probability that no set of k vertices has all of its connecting edges the same color. But this conclusion implies that there is at least one way of coloring the edges for which no set of k vertices has all of its connecting edges the same color.

Remarks

a. Whereas the preceding argument established a condition on n and k that guarantees the existence of a coloring scheme satisfying the desired property, it gives no information about how to obtain such a scheme (although one possibility would be simply to choose the colors at random, check to see if the resulting coloring satisfies the property, and repeat the procedure until it does).
b. The method of introducing probability into a problem whose statement is purely deterministic has been called the *probabilistic method*.† Other examples of this method are given in Theoretical **Exercise 24** and Examples 2t and 2u of **Chapter 7**.

†See N. Alon, J. Spencer, and P. Erdos, *The Probabilistic Method* (New York: John Wiley & Sons, Inc., 1992).